动态随机一般均衡模型：理论与应用
第 10 章：抵押融资约束

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A 动态系统

A.1 房地产模型

A.1.1 住房不作为生产要素（情形一）

无抵押融资约束

$$\beta\frac{C_t}{\mathbb{E}_t C_{t+1}}=\frac{1}{\hat{R}_t}$$
$$C_t N_t^{\varphi}=\mu A_t$$
$$\frac{\mathbb{E}_t C_{t+1}}{\hat{Q}_t H}=\beta\Big[\hat{R}_t-(1-\delta_h)\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}\Big]$$
$$Y_t=A_t N_t$$
$$Y_t=C_t$$
$$a_t=\rho_a a_{t-1}+\eta_t^a$$

有抵押融资约束

$$\beta^S\frac{C_t^S}{\mathbb{E}_t C_{t+1}^S}=\frac{1}{\hat{R}_t}$$
$$C_t^S(N_t^S)^{\varphi}=\hat{W}_t$$
$$C_t^B(N_t^B)^{\varphi}=\hat{W}_t$$
$$\frac{\mathbb{E}_t C_{t+1}^S}{\hat{Q}_t H_t^S}=\beta^S\Big[\hat{R}_t-(1-\delta_h)\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}\Big]$$
$$\hat{R}_t\hat{L}_t=\omega\mathbb{E}_t\hat{Q}_{t+1}H_t^B$$
$$\frac{\mathbb{E}_t C_{t+1}^B}{\hat{Q}_t H_t^B}=\frac{\mathbb{E}_t C_{t+1}^B}{C_t^B}-(1-\delta_h)\beta^B\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}-\omega\frac{\mathbb{E}_t C_{t+1}^B}{C_t^B}\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}\frac{1}{\hat{R}_t}+\beta^B\omega\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}$$
$$C_t^B+\hat{Q}_t H_t^B+\hat{R}_{t-1}\hat{L}_{t-1}=\mu A_t N_t^B+(1-\delta_h)\hat{Q}_t H_{t-1}^B+\hat{L}_t$$
$$Y_t=C_t^S+C_t^B$$
$$Y_t=A_t(N_t^S+N_t^B)$$
$$\hat{W}_t=\mu A_t$$
$$H=H_t^S+H_t^B$$
$$a_t=\rho_a a_{t-1}+\eta_t^a$$
$$\hat{\omega}_t=\rho_\omega\hat{\omega}_{t-1}+\eta_t^\omega$$

A.1.2 住房作为生产要素（情形二）

无抵押融资约束

$$\beta\frac{C_t}{\mathbb{E}_t C_{t+1}}=\frac{1}{\hat{R}_t}$$
$$C_t N_t^{\varphi}=\hat{W}_t$$
$$\frac{\mathbb{E}_t C_{t+1}}{\hat{Q}_t H_t^H}=\beta\Big[\hat{R}_t-(1-\delta_h)\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}\Big]$$
$$Y_t=A_t(H_t^F)^{\alpha}N_t^{1-\alpha}$$
$$N_t=\frac{(1-\alpha)\mu Y_t}{\hat{W}_t}$$
$$1=\frac{\alpha\mu Y_t}{\hat{Q}_t H_t^F}+\frac{(1-\delta_h)\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t\hat{R}_t}$$
$$Y_t=C_t$$
$$H=H_t^H+H_t^F$$
$$a_t=\rho_a a_{t-1}+\eta_t^a$$

有抵押融资约束

$$\beta^S\frac{C_t^S}{\mathbb{E}_t C_{t+1}^S}=\frac{1}{\hat{R}_t}$$
$$C_t^S(N_t^S)^{\varphi}=\hat{W}_t$$
$$C_t^B(N_t^B)^{\varphi}=\hat{W}_t$$
$$\frac{\mathbb{E}_t C_{t+1}^S}{\hat{Q}_t H_t^S}=\beta^S\Big[\hat{R}_t-(1-\delta_h)\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}\Big]$$
$$\hat{R}_t\hat{L}_t=\omega\mathbb{E}_t\hat{Q}_{t+1}H_t^B$$
$$\frac{\mathbb{E}_t C_{t+1}^B}{\hat{Q}_t H_t^B}=\frac{\mathbb{E}_t C_{t+1}^B}{C_t^B}-(1-\delta_h)\beta^B\mathbb{E}_t\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}-\omega\frac{\mathbb{E}_t C_{t+1}^B}{C_t^B}\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}\frac{1}{\hat{R}_t}+\beta^B\omega\frac{\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t}$$
$$C_t^B+\hat{Q}_t H_t^B+\hat{R}_{t-1}\hat{L}_{t-1}=\hat{W}_t N_t^B+(1-\delta_h)\hat{Q}_t H_{t-1}^B+\hat{L}_t$$
$$Y_t=C_t^S+C_t^B$$
$$Y_t=A_t(H_t^F)^{\alpha}N_t^{1-\alpha}$$
$$\hat{W}_t N_t=(1-\alpha)\mu Y_t$$
$$1=\frac{\alpha\mu Y_t}{\hat{Q}_t H_t^F}+\frac{(1-\delta_h)\mathbb{E}_t\hat{Q}_{t+1}}{\hat{Q}_t\hat{R}_t}$$
$$N_t=N_t^S+N_t^B$$
$$H=H_t^S+H_t^B+H_t^F$$
$$a_t=\rho_a a_{t-1}+\eta_t^a$$
$$\hat{\omega}_t=\rho_\omega\hat{\omega}_{t-1}+\eta_t^{\omega}$$

B 求解模型稳态

B.1 稳态系统

B.1.1 住房不作为生产要素（情形一）

设稳态时 $A=1$，以及：

$$\beta^S = \frac{1}{\hat{R}}$$
$$C^S(N^S)^{\varphi} = \mu$$
$$C^B(N^B)^{\varphi} = \mu$$
$$\frac{C^S}{\hat{Q}H^S} = \Omega_1$$
$$\hat{R}\hat{L} = \omega\hat{Q}H^B$$
$$C^B + \hat{Q}H^B + \hat{R}\hat{L} = \mu N^B + (1-\delta_h)\hat{Q}H^B + \hat{L}$$
$$\frac{C^B}{\hat{Q}H^B} = \Omega_2$$
$$Y = N^S + N^B$$
$$Y = C^S + C^B$$
$$H = H^S + H^B$$

其中，$\Omega_1 = 1-\beta^S(1-\delta_h)$，$\Omega_2 = 1-\beta^B(1-\delta_h)-\omega(\beta^S-\beta^B)$。

B.1.2 住房作为生产要素（情形二）

设稳态时 $A=1$，以及：

$$\beta^S = 1/\hat{R}$$
$$C^S(N^S)^{\varphi} = \hat{W}$$
$$C^B(N^B)^{\varphi} = \hat{W}$$
$$\frac{C^S}{\hat{Q}H^S} = \Omega_1$$
$$\hat{R}\hat{L} = \omega\hat{Q}H^B$$
$$\frac{C^B}{\hat{Q}H^B} = \Omega_2$$
$$C^B + \hat{Q}H^B + \hat{R}\hat{L} = \hat{W}N^B + (1-\delta_h)\hat{Q}H^B + \hat{L}$$
$$Y = (H^F)^{\alpha}N^{1-\alpha}$$
$$N = \frac{(1-\alpha)\mu}{\hat{W}}Y$$
$$1 = \frac{\alpha\mu Y}{\hat{Q}H^F} + \beta^S(1-\delta_h)$$
$$Y = C^S + C^B$$
$$N = N^S + N^B$$
$$H = H^S + H^B + H^F$$

其中：$\Omega_1 = 1-\beta^S(1-\delta_h)$，$\Omega_2 = 1-\beta^B(1-\delta_h)-\omega(\beta^S-\beta^B)$。

B.2 稳态均衡

B.2.1 住房不作为生产要素（情形一）

校准 H 使得 $\hat{Q}=1$。令：$\Omega^S=1+\big[\delta_h+\omega(1-\beta^S)\big]/\Omega_2$，可解得

$$\beta^S=\frac{1}{\hat{R}}$$
$$H^B=\mu(\Omega^S)^{-\frac{\varphi}{1+\varphi}}/\Omega_2$$
$$C^B=\mu(\Omega^S)^{-\frac{\varphi}{1+\varphi}}$$
$$N^B=(\Omega^S)^{\frac{1}{1+\varphi}}$$
$$\hat{L}=\omega\beta^S\mu(\Omega^S)^{-\frac{\varphi}{1+\varphi}}/\Omega_2$$

基于上述解以及其他方程化简得到关于 H 的方程：

$$\Big(\frac{\mu}{\Omega_1}\Big)^{\frac{1}{\varphi}}\left[H-\frac{\mu}{\Omega_2}(\Omega^S)^{-\frac{\varphi}{1+\varphi}}\right]^{-\frac{1}{\varphi}}=\Omega_1 H-\Big(\frac{\Omega_1}{\Omega_2}-1\Big)\mu(\Omega^S)^{-\frac{\varphi}{1+\varphi}}-(\Omega^S)^{\frac{1}{1+\varphi}}$$

可通过数值方法求得稳态 H 的值，进而可获得模型稳态解：

$$H^S=H-H^B,\quad C^S=\Omega_1 H^S,\quad N^S=\Big(\frac{\mu}{\Omega_1}\Big)^{\frac{1}{\varphi}}(H^S)^{-\frac{1}{\varphi}},\quad Y=C^S+C^B$$

B.2.2 住房作为生产要素（情形二）

校准稳态值 H 使得均衡时 $\hat{Q}=1$，稳态时，

$$\beta^S=1/\hat{R}$$
$$\hat{W}=(1-\alpha)\mu^{\frac{1}{1-\alpha}}\Big(\frac{\alpha}{\Omega_1}\Big)^{\frac{\alpha}{1-\alpha}}$$
$$H^B=\hat{W}(\Omega^S)^{-\frac{\varphi}{1+\varphi}}/\Omega_2$$
$$C^B=\hat{W}(\Omega^S)^{-\frac{\varphi}{1+\varphi}}$$
$$N^B=(\Omega^S)^{\frac{1}{1+\varphi}}$$
$$\hat{L}=\omega\beta^S\hat{W}(\Omega^S)^{-\frac{\varphi}{1+\varphi}}/\Omega_2$$

给定 H 有 $H^F=\frac{\alpha\mu}{1+\alpha\mu}\big(H-H^B+\frac{C^B}{\Omega_1}\big)$，$C^S=\Omega_1(H-H^B-H^F)$，最后：

$$\hat{W}=C^S\left[\frac{(1-\alpha)\mu}{\hat{W}}(C^S+C^B)-N^B\right]^{\varphi}$$

可基于上述方程通过数值方法求得稳态 H 的值，进而可获得模型稳态解：

$$H^F=\frac{\alpha\mu}{1+\alpha\mu}\Big(H-H^B-\frac{C^B}{\Omega_1}\Big),\quad H^S=H-H^B-H^F,\quad C^S=\Omega_1 H^S$$
$$Y=C^S+C^B,\quad N=(1-\alpha)\mu Y/\hat{W},\quad N^S=N-N^B$$

C 资本作为抵押品模型

C.1 动态系统

C.1.1 基准模型（资本不作为抵押品）

$$\beta\hat{R}_t C_t=\mathbb{E}_t C_{t+1}$$

$$C_t N_t^{\varphi} = \hat{W}_t$$
$$(1-\alpha) Y_t = \hat{W}_t N_t$$
$$K_{t+1} = (1-\delta) K_t + I_t$$
$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}$$
$$\hat{Q}_t = 1 + \theta_I \Big[\frac{1}{2} \Big(\frac{I_t}{I_{t-1}} - 1 \Big)^2 + \Big(\frac{I_t}{I_{t-1}} - 1 \Big) \frac{I_t}{I_{t-1}} - \mathbb{E}_t \Lambda_{t,t+1} \Big(\frac{I_{t+1}}{I_t} - 1 \Big) \Big(\frac{I_{t+1}}{I_t} \Big)^2 \Big]$$
$$\hat{R}_t = \mathbb{E}_t \Big[(1-\delta) \frac{\hat{Q}_{t+1}}{\hat{Q}_t} + \alpha \frac{Y_{t+1}}{\hat{Q}_t K_{t+1}} \Big]$$
$$Y_t = C_t + \Big[1 + \frac{\theta_I}{2} \Big(\frac{I_t}{I_{t-1}} - 1 \Big)^2 \Big] I_t$$

C.1.2 资本作为抵押品

$$\beta \hat{R}_t C_t = \mathbb{E}_t C_{t+1}$$
$$C_t N_t^{\varphi} = \hat{W}_t$$
$$(1-\alpha) Y_t = \hat{W}_t N_t$$
$$K_{t+1} = (1-\delta) K_t + I_t$$
$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}$$
$$\hat{Q}_t = 1 + \theta_I \Big[\frac{1}{2} \Big(\frac{I_t}{I_{t-1}} - 1 \Big)^2 + \Big(\frac{I_t}{I_{t-1}} - 1 \Big) \frac{I_t}{I_{t-1}} - \mathbb{E}_t \Lambda_{t,t+1} \Big(\frac{I_{t+1}}{I_t} - 1 \Big) \Big(\frac{I_{t+1}}{I_t} \Big)^2 \Big]$$
$$\hat{Q}_t = \beta^E \mathbb{E}_t \frac{C_t^E}{C_{t+1}^E} \Big[(1-\delta-\omega) \hat{Q}_{t+1} + \alpha \frac{Y_{t+1}}{K_{t+1}} \Big] + \omega \mathbb{E}_t \frac{\hat{Q}_{t+1}}{\hat{R}_t}$$
$$\hat{R}_t \hat{L}_t = \omega \mathbb{E}_t (\hat{Q}_{t+1} K_{t+1})$$
$$C_t^E + \hat{Q}_t I_t + \hat{W}_t N_t + \hat{R}_{t-1} \hat{L}_{t-1} = Y_t + \hat{L}_t$$
$$Y_t = C_t + C_t^E + \Big[1 + \frac{\theta_I}{2} \Big(\frac{I_t}{I_{t-1}} - 1 \Big)^2 \Big] I_t$$
$$a_t = \rho_a a_{t-1} + \eta_t^a$$

D 求解模型稳态

D.1 稳态系统

设稳态时 $A = 1$。稳态时，$\hat{R} = \frac{1}{\beta}$，$\hat{Q} = 1$，以及：

$$C N^{\varphi} = \hat{W}$$
$$(1-\alpha) Y = \hat{W} N$$
$$\beta^E \Big(1 - \delta + \alpha \frac{Y}{K} \Big) + \omega C^E \lambda^M = 1$$
$$\beta^E \hat{R} = 1 - C^E \lambda^M \hat{R}$$
$$Y = K^{\alpha} N^{1-\alpha}$$
$$I = \delta K$$
$$\hat{R} \hat{L} = \omega K$$
$$Y = C + C^E + I$$

$$C^E + I + \hat{W}N + \hat{R}\hat{L} = Y + \hat{L}$$

D.2 稳态均衡

$$\hat{W} = (1-\alpha)\big(\frac{\beta^E\alpha}{\Omega_3}\big)^{\frac{\alpha}{1-\alpha}}$$

$$Y = \Big(\frac{1-\alpha}{1-\alpha+\alpha\omega(1-\beta)\beta^E/\Omega_3}\Big)^{\frac{1}{1+\varphi}}\Big(\frac{\alpha\beta^E}{\Omega_3}\Big)^{\frac{\alpha}{1-\alpha}}$$

$$N = \frac{1-\alpha}{\hat{W}}Y$$

$$K = \frac{\beta^E\alpha}{\Omega_3}Y$$

$$I = \delta K$$

$$C = (1-\alpha)^{-\varphi}\hat{W}^{1+\varphi}Y^{-\varphi}$$

$$C^E = \alpha\Big\{1-\big[\delta+\omega(1-\beta)\big]\frac{\beta^E}{\Omega_3}\Big\}Y$$

$$\hat{L} = \beta\omega K$$

其中 $\Omega_3 = 1-\omega(\beta-\beta^E)-\beta^E(1-\delta)$。

对于基准模型，稳态均衡为：

$$\hat{W} = (1-\alpha)\Big(\frac{\alpha\beta}{1-\beta(1-\delta)}\Big)^{\frac{\alpha}{1-\alpha}}$$

$$Y = \Big[\frac{(1-\alpha)[1-\beta(1-\delta)]}{1-\beta(1-\delta)-\alpha\beta\delta}\Big]^{\frac{1}{1+\varphi}}\Big(\frac{\alpha\beta}{1-\beta(1-\delta)}\Big)^{\frac{\alpha}{1-\alpha}}$$

$$N = \frac{1-\alpha}{\hat{W}}Y$$

$$K = \frac{\alpha\beta}{1-\beta(1-\delta)}Y$$

$$I = \delta K$$

$$C = (1-\alpha)^{-\varphi}\hat{W}^{1+\varphi}Y^{-\varphi}$$